



UNIVERSITY OF ENGINEERING AND TECHNOLOGY MARDAN

Department: Computer Sceince

Course: Mobile App Develepment

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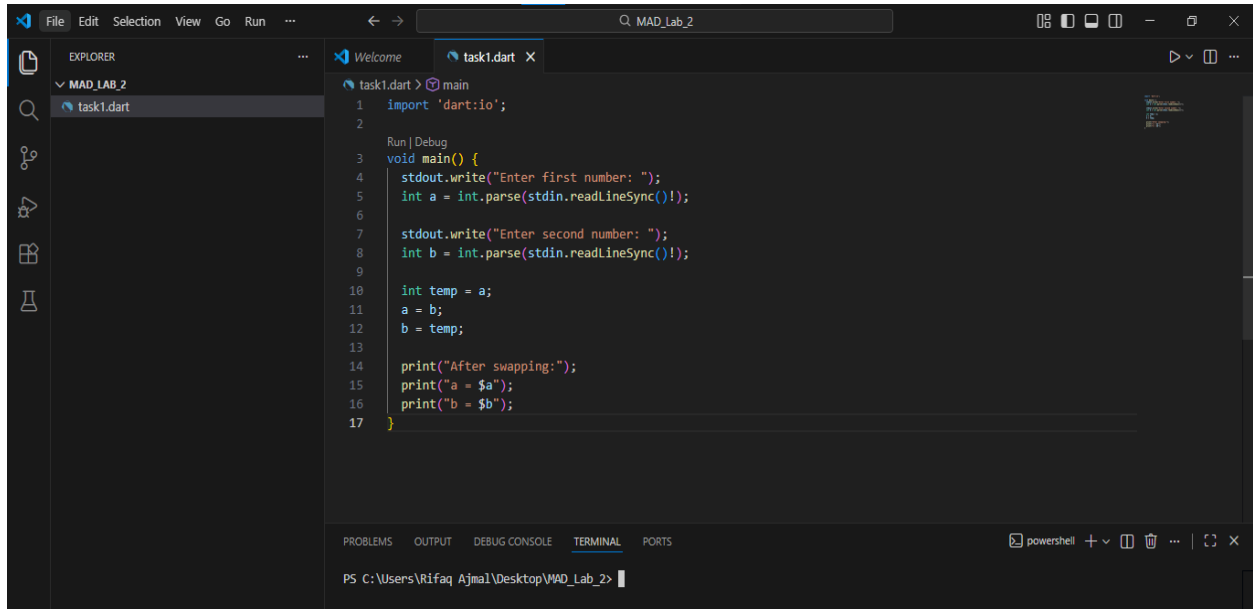
Stream: Computer Science (Section A)

Date : 3 Feb 2026

Lab : 2

1. Swapping of two number using third var?

Code

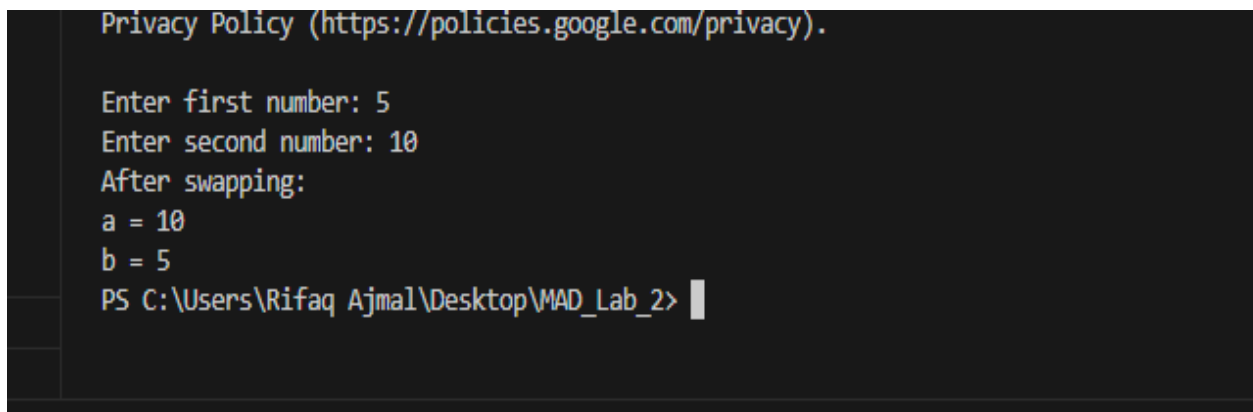


The screenshot shows an IDE with a file explorer on the left containing a folder 'MAD_Lab_2' and a file 'task1.dart'. The main editor displays the following Dart code:

```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter first number: ");
5   int a = int.parse(stdin.readLineSync());
6
7   stdout.write("Enter second number: ");
8   int b = int.parse(stdin.readLineSync());
9
10  int temp = a;
11  a = b;
12  b = temp;
13
14  print("After swapping:");
15  print("a = $a");
16  print("b = $b");
17 }
```

The bottom of the IDE shows a terminal window with the command prompt 'PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2>'.

Output



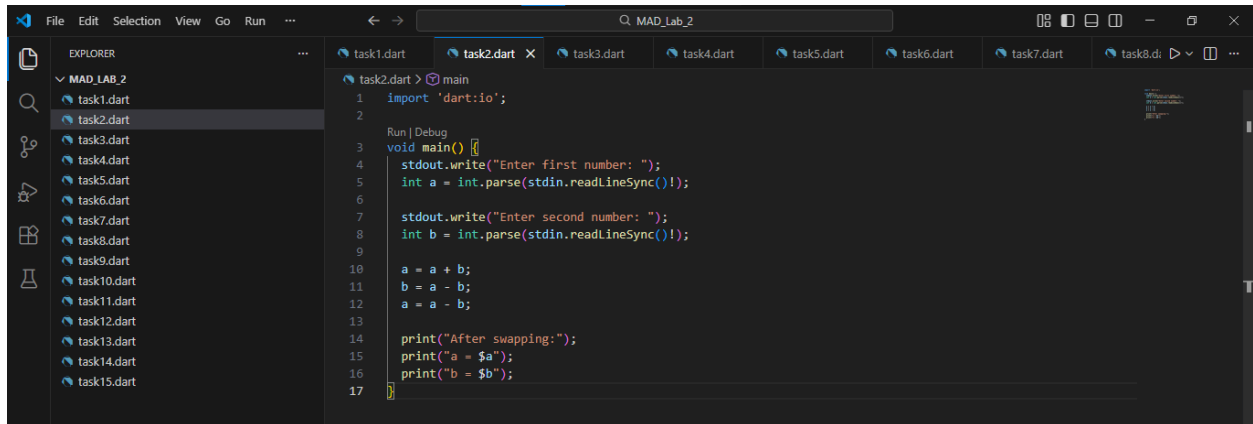
The screenshot shows a terminal window with the following output:

```
Privacy Policy (https://policies.google.com/privacy).

Enter first number: 5
Enter second number: 10
After swapping:
a = 10
b = 5
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2>
```

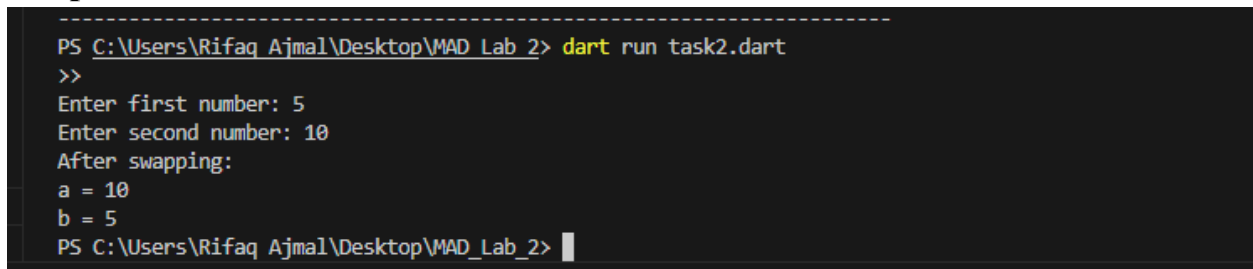
2. Swapping of two number without third variable?

Code



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter first number: ");
5   int a = int.parse(stdin.readLineSync());
6
7   stdout.write("Enter second number: ");
8   int b = int.parse(stdin.readLineSync());
9
10  a = a + b;
11  b = a - b;
12  a = a - b;
13
14  print("After swapping:");
15  print("a = $a");
16  print("b = $b");
17 }
```

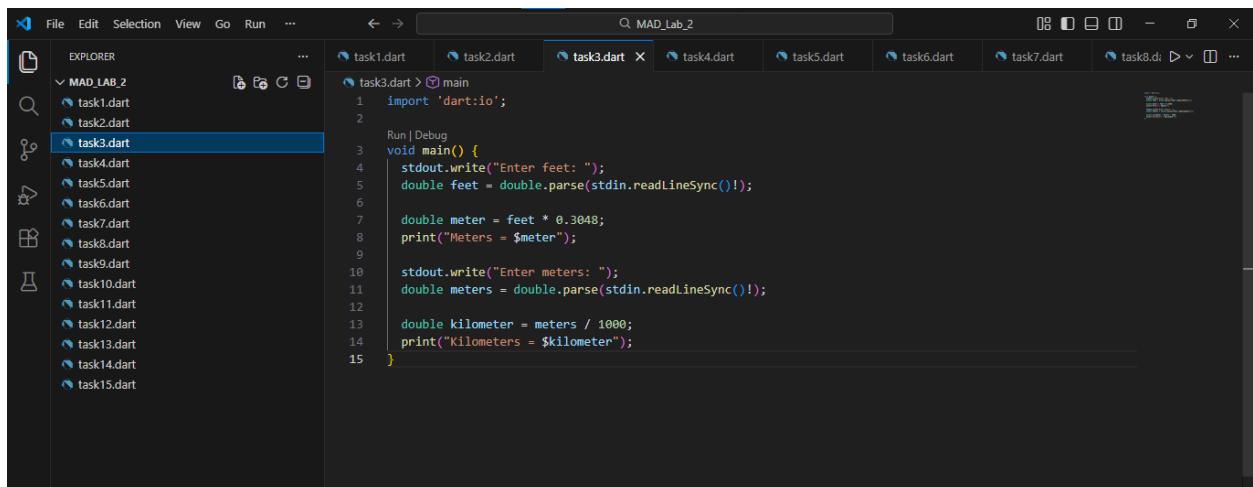
Output



```
PS C:\Users\Rifag Ajmal\Desktop\MAD_Lab_2> dart run task2.dart
>>
Enter first number: 5
Enter second number: 10
After swapping:
a = 10
b = 5
PS C:\Users\Rifag Ajmal\Desktop\MAD_Lab_2>
```

3. Convert feet to meter and meter to kilometer.

Code



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter feet: ");
5   double feet = double.parse(stdin.readLineSync());
6
7   double meter = feet * 0.3048;
8   print("Meters = $meter");
9
10  stdout.write("Enter meters: ");
11  double meters = double.parse(stdin.readLineSync());
12
13  double kilometer = meters / 1000;
14  print("Kilometers = $kilometer");
15 }
```

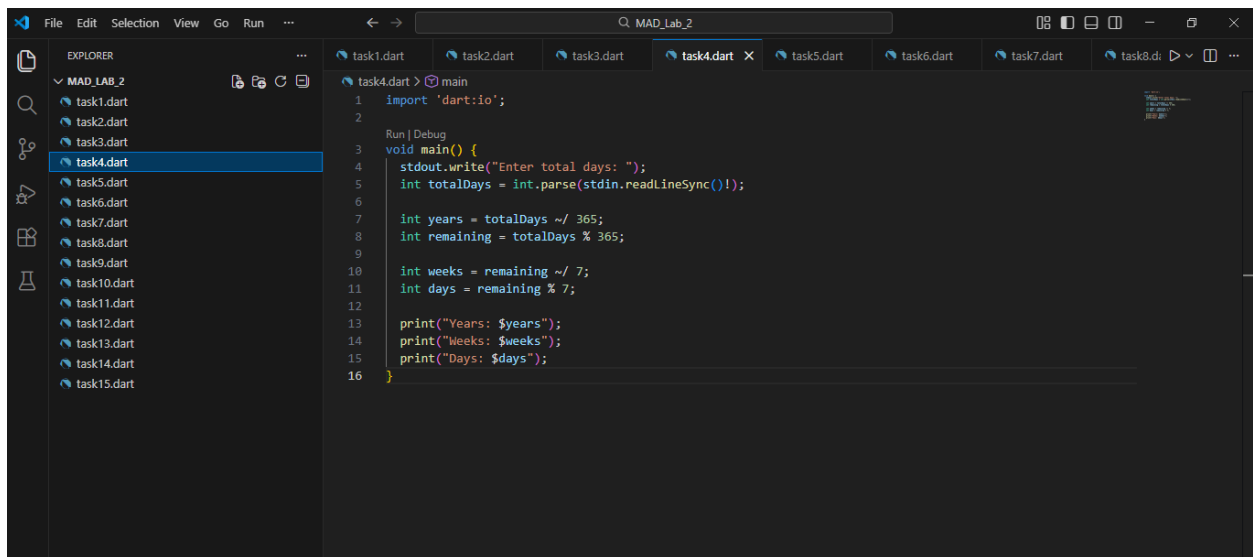
Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaa Ajmal\Desktop\MAD_Lab_2> dart run task3.dart
>>
Enter feet: 6
Meters = 1.8288000000000002
Enter meters: |
```

4. Write a Dart program that prompts the user to enter a total number of days as an integer. The program should then convert and display that value into its equivalent breakdown of Years, Weeks, and Days.

Code



```
File Edit Selection View Go Run ...  MAD_Lab_2
task1.dart task2.dart task3.dart task4.dart task5.dart task6.dart task7.dart task8.d...
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter total days: ");
5   int totalDays = int.parse(stdin.readLineSync()!);
6
7   int years = totalDays ~/ 365;
8   int remaining = totalDays % 365;
9
10  int weeks = remaining ~/ 7;
11  int days = remaining % 7;
12
13  print("Years: $years");
14  print("Weeks: $weeks");
15  print("Days: $days");
16 }
```

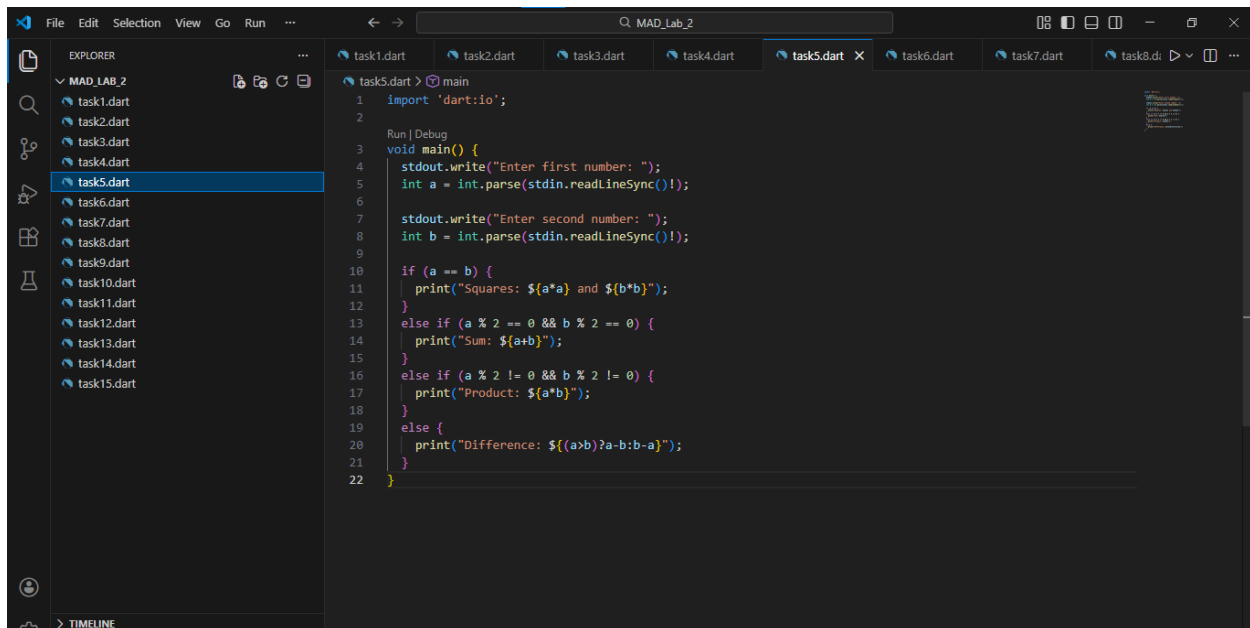
Output

```
Enter total days: 890
Years: 2
Weeks: 22
Days: 6
PS C:\Users\Rifaa Ajmal\Desktop\MAD_Lab_2> |
```

5. Write a Dart program that takes input from the user for two numbers and performs the following operations:

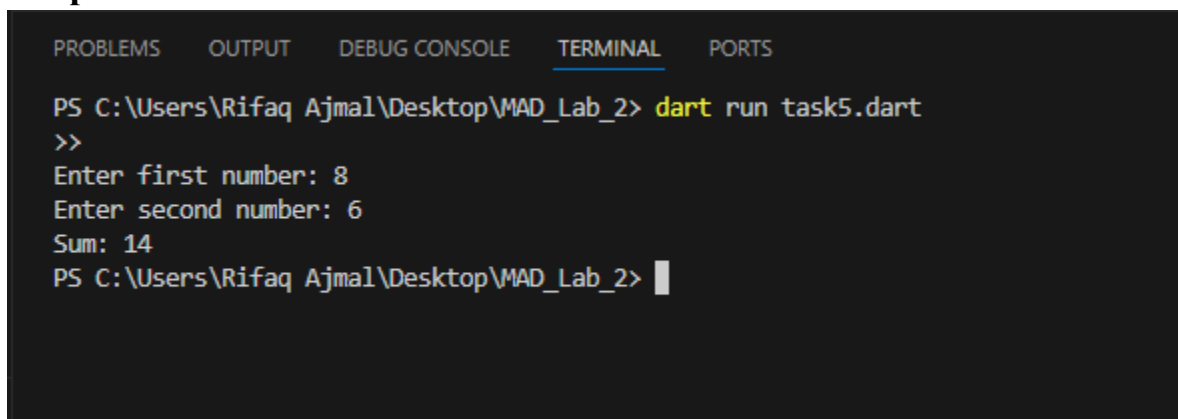
- a) Check if both numbers are even, if so, print their sum.
- b) If both numbers are odd, print their product.
- c) If one number is even and the other is odd, print the difference (larger number minus the smaller number).
- d) If both numbers are equal, print their squares.

Code



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter first number: ");
5   int a = int.parse(stdin.readLineSync());
6
7   stdout.write("Enter second number: ");
8   int b = int.parse(stdin.readLineSync());
9
10  if (a == b) {
11    print("Squares: ${a*a} and ${b*b}");
12  }
13  else if (a % 2 == 0 && b % 2 == 0) {
14    print("Sum: ${a+b}");
15  }
16  else if (a % 2 != 0 && b % 2 != 0) {
17    print("Product: ${a*b}");
18  }
19  else {
20    print("Difference: ${((a>b)?a-b:b-a)}");
21  }
22 }
```

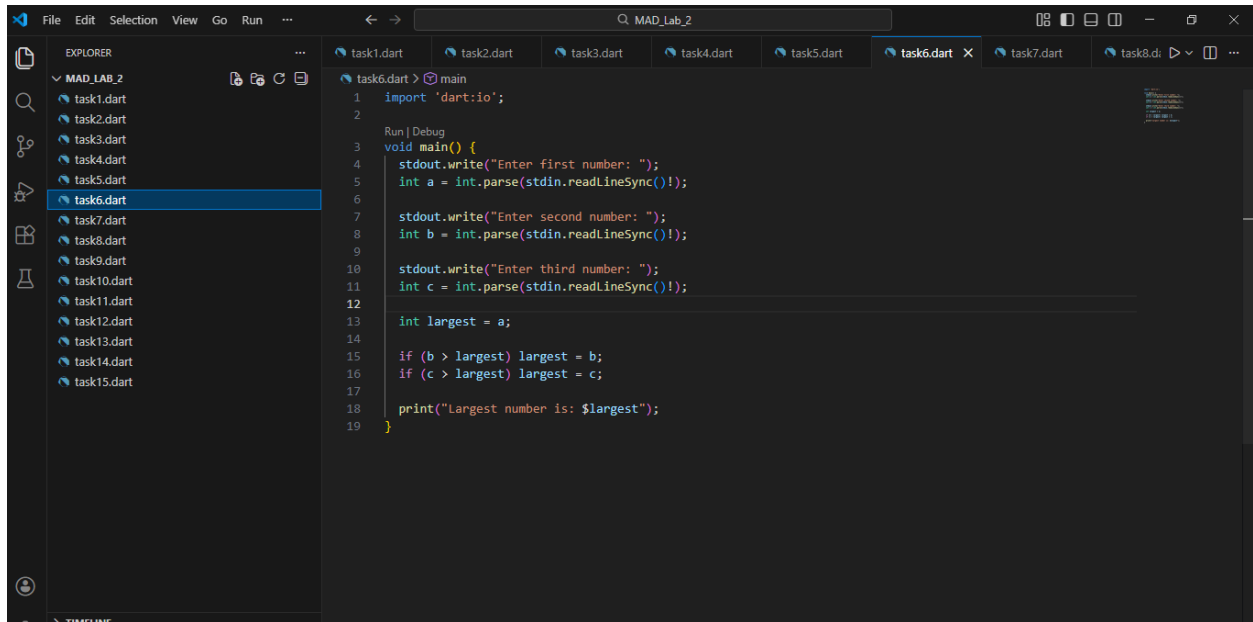
Output



```
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task5.dart
>>
Enter first number: 8
Enter second number: 6
Sum: 14
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```

6. Write a Dart program that takes three numbers from the user and finds the largest among them. Print the largest number.

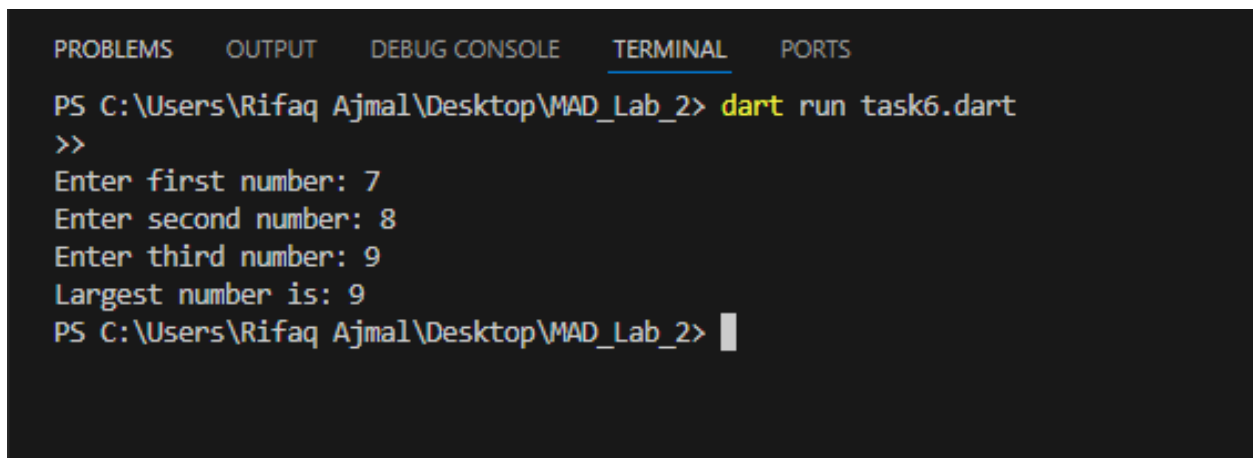
Code



The screenshot shows an IDE with a file explorer on the left containing a folder named 'MAD_LAB_2' with files task1.dart through task15.dart. The main editor displays 'task6.dart' with the following Dart code:

```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter first number: ");
5   int a = int.parse(stdin.readLineSync()!);
6
7   stdout.write("Enter second number: ");
8   int b = int.parse(stdin.readLineSync()!);
9
10  stdout.write("Enter third number: ");
11  int c = int.parse(stdin.readLineSync()!);
12
13  int largest = a;
14
15  if (b > largest) largest = b;
16  if (c > largest) largest = c;
17
18  print("Largest number is: $largest");
19 }
```

Output

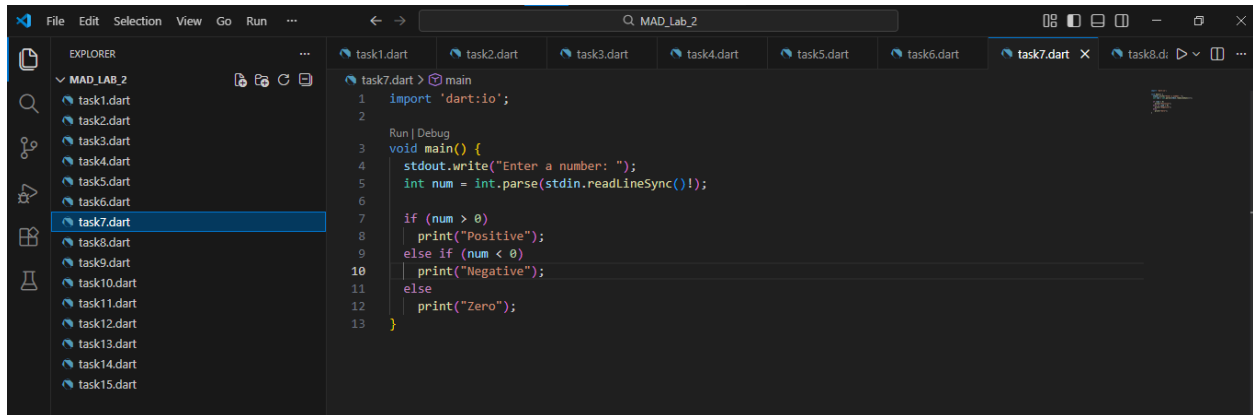


The screenshot shows a terminal window with the following output:

```
PS C:\Users\Rifaa Ajmal\Desktop\MAD_Lab_2> dart run task6.dart
>>
Enter first number: 7
Enter second number: 8
Enter third number: 9
Largest number is: 9
PS C:\Users\Rifaa Ajmal\Desktop\MAD_Lab_2> |
```

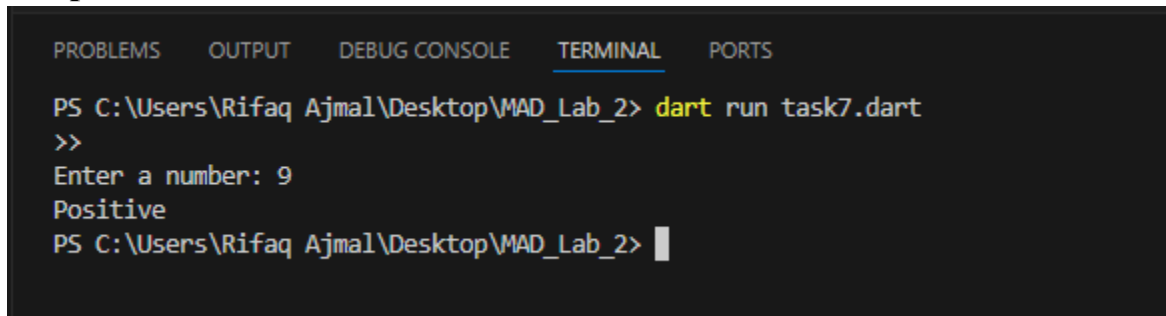
7. Write a Dart program that takes a number from the user and checks if it's positive, negative, or zero. Print "Positive" if it's positive, "Negative" if it's negative, and "Zero" if it's zero.

Code



```
task7.dart > main
1 import 'dart:io';
2
3 Run | Debug
4 void main() {
5     stdout.write("Enter a number: ");
6     int num = int.parse(stdin.readLineSync()!);
7
8     if (num > 0)
9         print("Positive");
10    else if (num < 0)
11        print("Negative");
12    else
13        print("Zero");
14 }
```

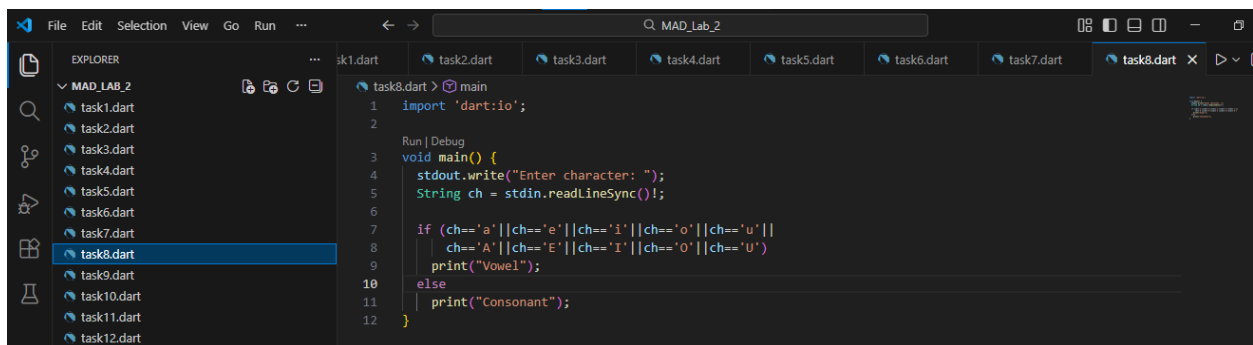
Output



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task7.dart
>>
Enter a number: 9
Positive
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```

8. Write a Dart program that takes a character from the user and checks if it's a vowel (a, e, i, o, u) or a consonant. Print "Vowel" if it's a vowel, otherwise print "Consonant".

Code



```
task8.dart > main
1 import 'dart:io';
2
3 Run | Debug
4 void main() {
5     stdout.write("Enter character: ");
6     String ch = stdin.readLineSync()!;
7
8     if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
9         ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U')
10        print("Vowel");
11    else
12        print("Consonant");
13 }
```

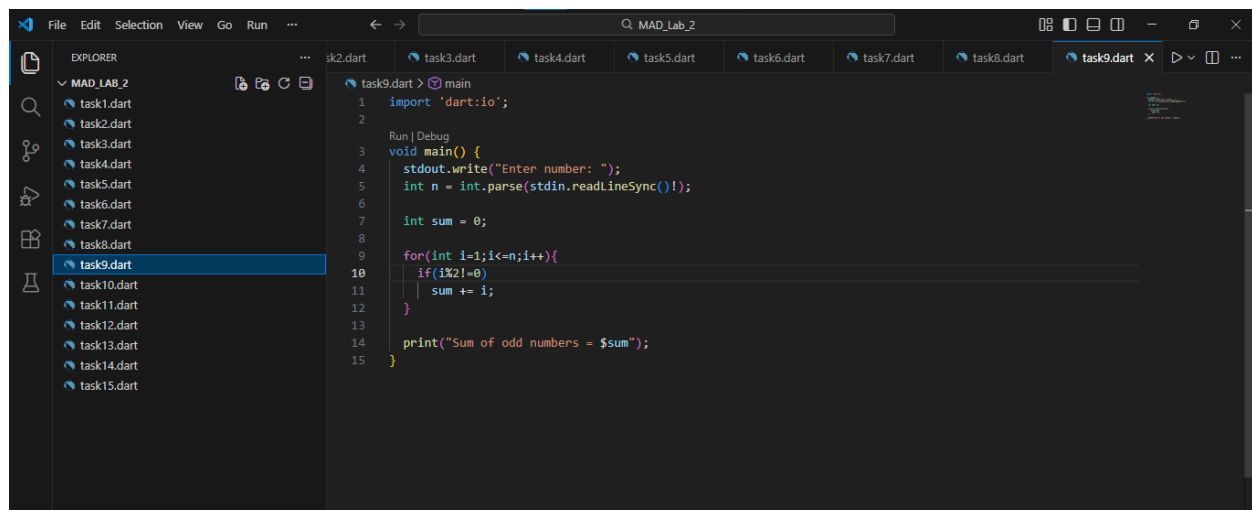
Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task8.dart
>>
Enter character: 7
Consonant
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```

9. Ask from user to enter the number and find sum of all odd numbers up to that number?

Code



```
File Edit Selection View Go Run ...  MAD_Lab_2
EXPLORER  task2.dart task3.dart task4.dart task5.dart task6.dart task7.dart task8.dart task9.dart
MAD LAB 2
task1.dart
task2.dart
task3.dart
task4.dart
task5.dart
task6.dart
task7.dart
task8.dart
task9.dart
task10.dart
task11.dart
task12.dart
task13.dart
task14.dart
task15.dart

task9.dart > main
1 import 'dart:io';
2
3 Run | Debug
4 void main() {
5     stdout.write("Enter number: ");
6     int n = int.parse(stdin.readLineSync()!);
7
8     int sum = 0;
9
10    for(int i=1; i<=n; i++){
11        if(i%2!=0){
12            sum += i;
13        }
14    }
15    print("Sum of odd numbers = $sum");
16 }
```

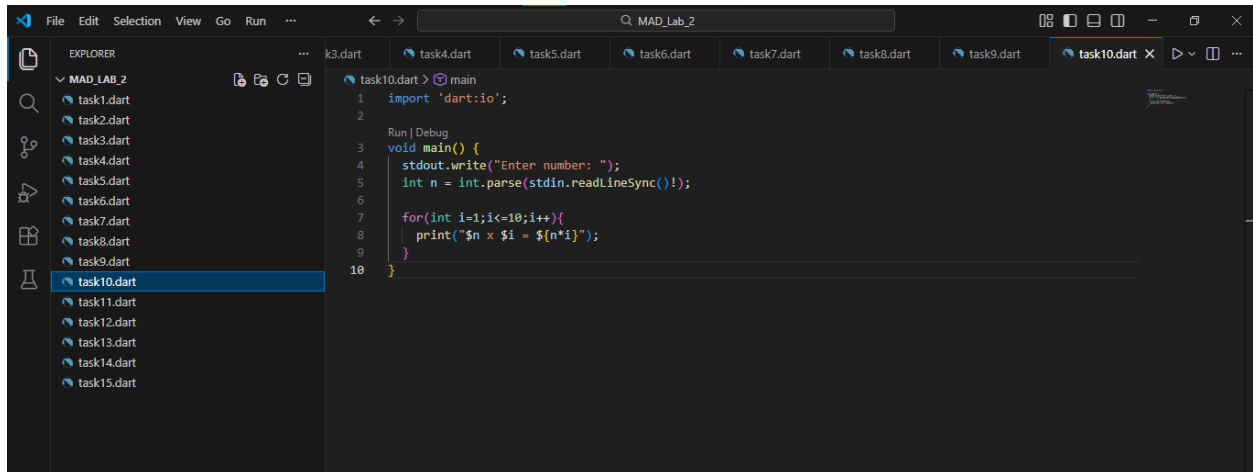
Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task9.dart
>>
Enter number: 8
Sum of odd numbers = 16
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```


10. Write a program that prompts the user to input a positive integer. It should then print the multiplication table of that number.

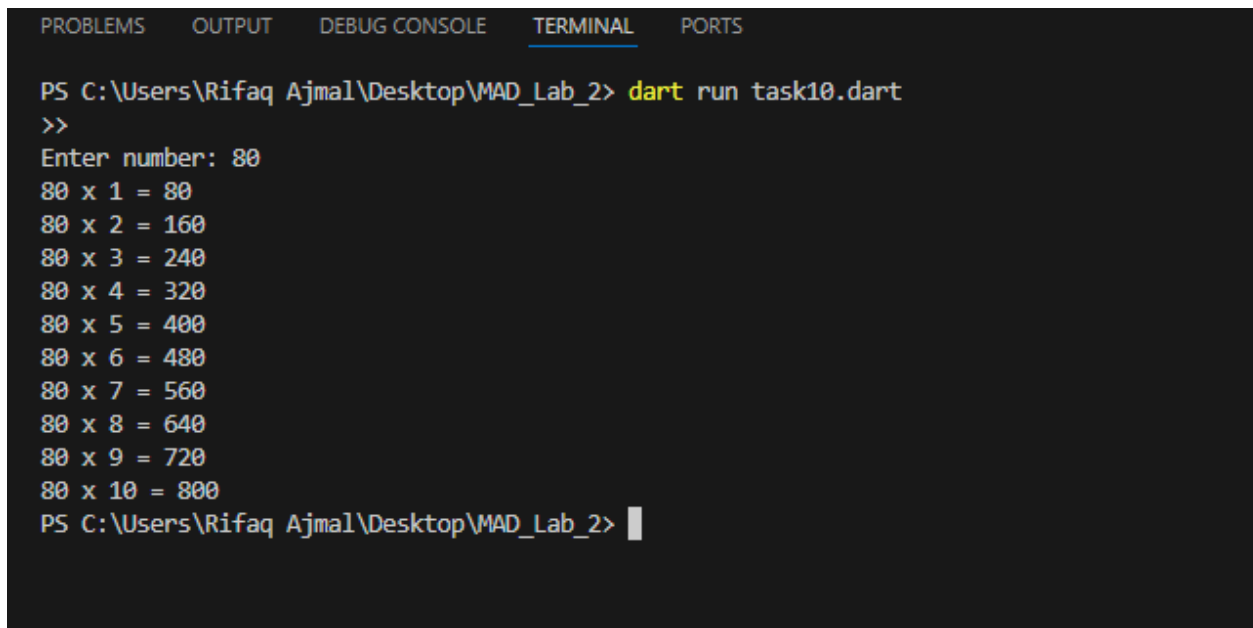
Code



The screenshot shows an IDE with a file explorer on the left containing a directory named 'MAD_Lab_2' with files task1.dart through task15.dart. The main editor displays the code for task10.dart:

```
task10.dart > main
1 import 'dart:io';
2
3 void main() {
4     stdout.write("Enter number: ");
5     int n = int.parse(stdin.readLineSync()!);
6
7     for(int i=1; i<=10; i++){
8         print("$n x $i = ${n*i}");
9     }
10 }
```

Output



The screenshot shows a terminal window with the following output:

```
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task10.dart
>>
Enter number: 80
80 x 1 = 80
80 x 2 = 160
80 x 3 = 240
80 x 4 = 320
80 x 5 = 400
80 x 6 = 480
80 x 7 = 560
80 x 8 = 640
80 x 9 = 720
80 x 10 = 800
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> |
```

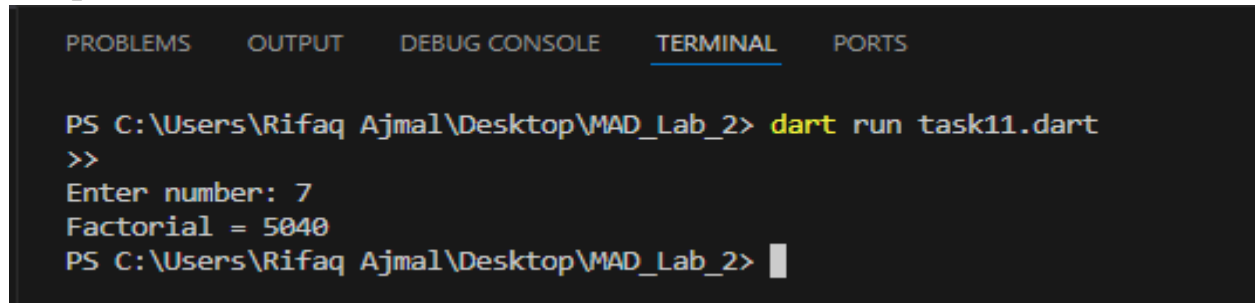
11. Write a program to find the factorial value of any number entered through the keyboard

Code



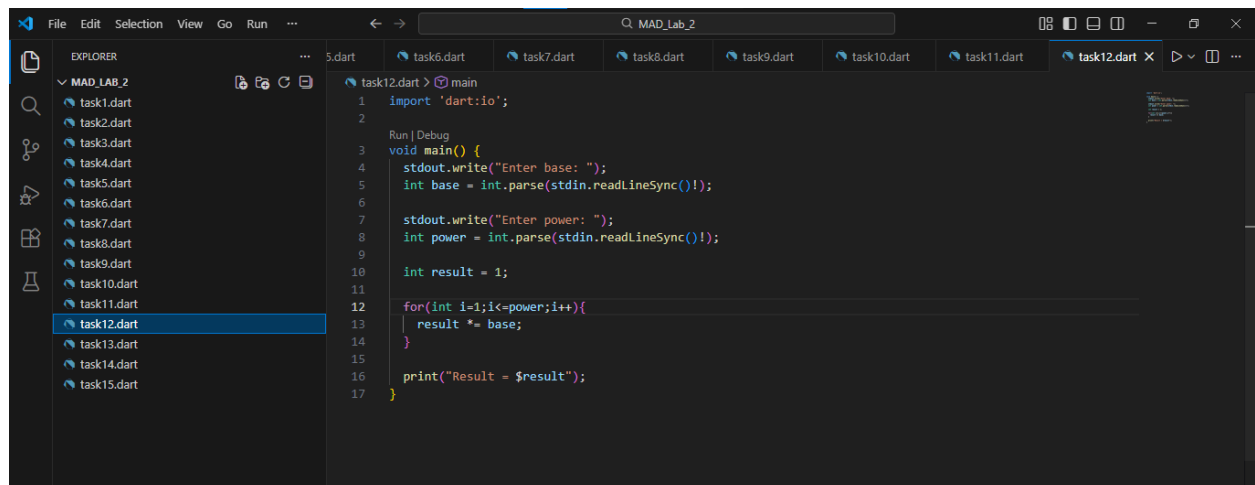
```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter number: ");
5   int n = int.parse(stdin.readLineSync()!);
6
7   int fact = 1;
8
9   for(int i=1; i<=n; i++){
10     fact *= i;
11   }
12
13   print("Factorial = $fact");
14 }
```

Output



```
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task11.dart
>>
Enter number: 7
Factorial = 5040
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```

12. Two numbers are entered through the keyboard. Write a program to find the value of one number raised to the power of another



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter base: ");
5   int base = int.parse(stdin.readLineSync()!);
6
7   stdout.write("Enter power: ");
8   int power = int.parse(stdin.readLineSync()!);
9
10  int result = 1;
11
12  for(int i=1; i<=power; i++){
13    result *= base;
14  }
15
16  print("Result = $result");
17 }
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task12.dart
>>
Enter base: 80
Enter power: 2
Result = 6400
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> |
```

13. Write a program that take prompt from user that how much number he want to enter and then find sum of all that numbers.

Code

```
File Edit Selection View Go Run ...  MAD_Lab_2
EXPLORER  task13.dart task7.dart task8.dart task9.dart task10.dart task11.dart task12.dart task13.dart
MAD_LAB_2
task1.dart
task2.dart
task3.dart
task4.dart
task5.dart
task6.dart
task7.dart
task8.dart
task9.dart
task10.dart
task11.dart
task12.dart
task13.dart
task14.dart
task15.dart

task13.dart <main>
1  import 'dart:io';
2
3  Run | Debug
4  void main() {
5      stdout.write("How many numbers? ");
6      int count = int.parse(stdin.readLineSync());
7
8      int sum = 0;
9
10     for(int i=1;i<=count;i++){
11         stdout.write("Enter number $i: ");
12         int num = int.parse(stdin.readLineSync());
13         sum += num;
14     }
15     print("Total sum = $sum");
16 }
```

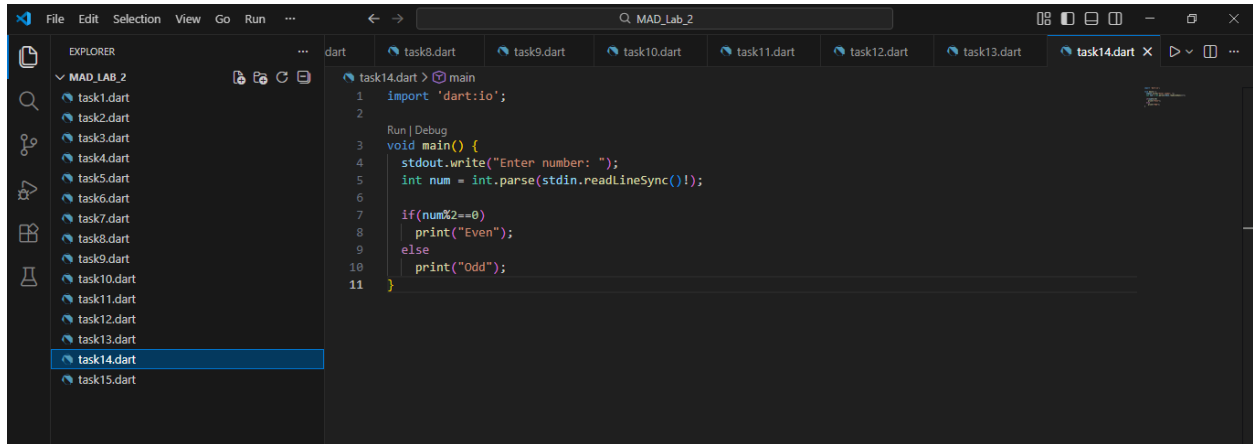
Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task13.dart
>>
How many numbers? 7
Enter number 1: 6
Enter number 2: 7
Enter number 3: 5
Enter number 4: 76
Enter number 5: 5
Enter number 6: 43
Enter number 7: 6
Total sum = 148
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> |
```

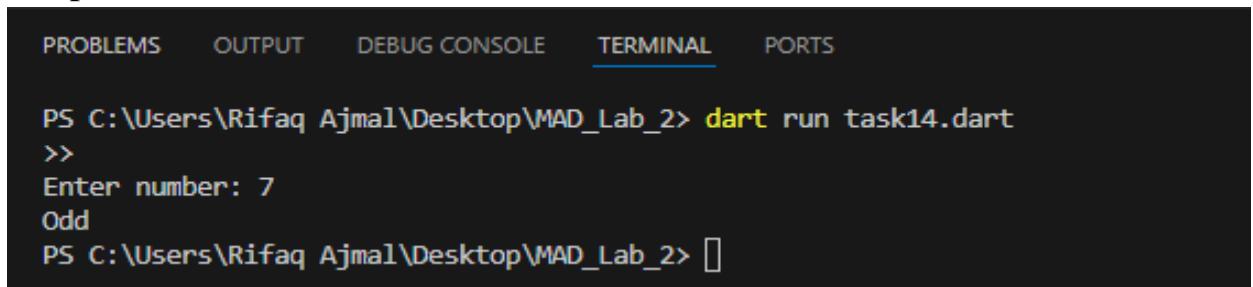
14. Write a program that prompts the user to input a positive integer. It should then output a message indicating whether the number is a even or odd number.

Code



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter number: ");
5   int num = int.parse(stdin.readLineSync()!);
6
7   if(num%2==0)
8     print("Even");
9   else
10    print("Odd");
11 }
```

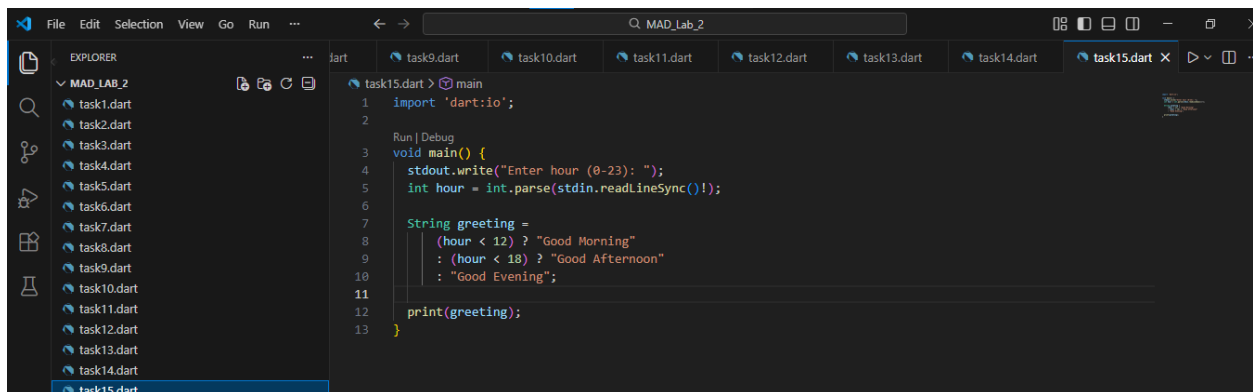
Output



```
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task14.dart
>>
Enter number: 7
Odd
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> 
```

15. Use the ternary operator to generate a custom greeting based on the time of the day.

Code



```
1 import 'dart:io';
2
3 void main() {
4   stdout.write("Enter hour (0-23): ");
5   int hour = int.parse(stdin.readLineSync()!);
6
7   String greeting =
8     (hour < 12) ? "Good Morning"
9     : (hour < 18) ? "Good Afternoon"
10     : "Good Evening";
11
12   print(greeting);
13 }
```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> dart run task15.dart  
>>  
Enter hour (0-23): 7  
Good Morning  
PS C:\Users\Rifaq Ajmal\Desktop\MAD_Lab_2> |
```